Hardening A Linux System

Part 1: System Updates and Patch Management

Check what updates are available:

Apply all available updates:

```
student@LinuxTargetVM3:~$ sudo apt update && apt list --upgradable
[sudo] password for student:
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease
Get:5 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1190 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [282 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1683 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [335 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1424 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [323 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.6 kB]
Fetched 5376 kB in 1s (3667 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Listing... Done
```

Write down what got updated, including anything related to the kernel.

```
student@LinuxTargetVM3:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

Check if a reboot is needed to finish applying patches:

If this file exists and prints a message, note it in your findings. Do not reboot the machine yourself. Flag it to your instructor first.

```
student@LinuxTargetVM3:~$ cat /var/run/reboot-required
*** System restart required ***
```

Install unattended-upgrades so future security patches apply automatically:

Confirm it's enabled:

```
student@LinuxTargetVM3:~$ systemctl status unattended-upgrades
● unattended-upgrades.service - Unattended Upgrades Shutdown
     Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-08-05 17:05:58 UTC; 6 days ago
       Docs: man:unattended-upgrade(8)
   Main PID: 1087 (unattended-upgr)
      Tasks: 2 (limit: 9434)
     Memory: 9.4M (peak: 9.5M)
        CPU: 131ms
     CGroup: /system.slice/unattended-upgrades.service
             └─1087 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

Aug 05 17:05:58 LinuxTargetVM3 systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
student@LinuxTargetVM3:~$
```

Part 2: Disabling Unused Services and Ports

Running services:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ systemctl list-units --type=service --state=running
  UNIT                           LOAD   ACTIVE SUB     DESCRIPTION
  auditd.service                 loaded active running Security Auditing Service
  chrony.service                 loaded active running chrony, an NTP client/server
  cron.service                   loaded active running Regular background program processing daemon
  dbus.service                   loaded active running D-Bus System Message Bus
  fail2ban.service               loaded active running Fail2Ban Service
  fwupd.service                  loaded active running Firmware update daemon
  getty@tty1.service             loaded active running Getty on tty1
  hv-kvp-daemon.service          loaded active running Hyper-V KVP Protocol Daemon
  inetutils-inetd.service        loaded active running GNU Network Utilities internet superserver
  ModemManager.service           loaded active running Modem Manager
  multipathd.service             loaded active running Device-Mapper Multipath Device Controller
  networkd-dispatcher.service    loaded active running Dispatcher daemon for systemd-networkd
  packagekit.service             loaded active running PackageKit Daemon
  polkit.service                 loaded active running Authorization Manager
  postfix@-.service              loaded active running Postfix Mail Transport Agent (instance -)
  rsyslog.service                loaded active running System Logging Service
  serial-getty@ttyS0.service     loaded active running Serial Getty on ttyS0
  ssh.service                    loaded active running OpenBSD Secure Shell server
  systemd-journald.service       loaded active running Journal Service
  systemd-logind.service         loaded active running User Login Management
  systemd-networkd.service       loaded active running Network Configuration
  systemd-resolved.service       loaded active running Network Name Resolution
  systemd-udevd.service          loaded active running Rule-based Manager for Device Events and Files
  udisks2.service                loaded active running Disk Manager
  unattended-upgrades.service    loaded active running Unattended Upgrades Shutdown
  user@1000.service              loaded active running User Manager for UID 1000
  walinuxagent.service           loaded active running Azure Linux Agent

lines 1-29...skipping...
  UNIT                           LOAD   ACTIVE SUB     DESCRIPTION
  auditd.service                 loaded active running Security Auditing Service
  chrony.service                 loaded active running chrony, an NTP client/server
  cron.service                   loaded active running Regular background program processing daemon
  dbus.service                   loaded active running D-Bus System Message Bus
  fail2ban.service               loaded active running Fail2Ban Service
  fwupd.service                  loaded active running Firmware update daemon
  getty@tty1.service             loaded active running Getty on tty1
  hv-kvp-daemon.service          loaded active running Hyper-V KVP Protocol Daemon
  inetutils-inetd.service        loaded active running GNU Network Utilities internet superserver
  ModemManager.service           loaded active running Modem Manager
  multipathd.service             loaded active running Device-Mapper Multipath Device Controller
  networkd-dispatcher.service    loaded active running Dispatcher daemon for systemd-networkd
  packagekit.service             loaded active running PackageKit Daemon
  polkit.service                 loaded active running Authorization Manager
```

```
  postfix@-.service            loaded active running Postfix Mail Transport Agent (instance -)
  rsyslog.service              loaded active running System Logging Service
  serial-getty@ttyS0.service   loaded active running Serial Getty on ttyS0
  ssh.service                  loaded active running OpenBSD Secure Shell server
  systemd-journald.service     loaded active running Journal Service
  systemd-logind.service       loaded active running User Login Management
  systemd-networkd.service     loaded active running Network Configuration
  systemd-resolved.service     loaded active running Network Name Resolution
  systemd-udevd.service        loaded active running Rule-based Manager for Device Events and Files
  udisks2.service              loaded active running Disk Manager
  unattended-upgrades.service loaded active running Unattended Upgrades Shutdown
  user@1000.service            loaded active running User Manager for UID 1000
  walinuxagent.service         loaded active running Azure Linux Agent

Legend: LOAD   → Reflects whether the unit definition was properly loaded.
        ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
        SUB    → The low-level unit activation state, values depend on unit type.

27 loaded units listed.
~
~
~
~
~
~
~
~
~
~
~
~
~
```

```
None
systemctl list-units --type=service --state=running
```

Listening ports:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo ss -tulpn
Netid    State     Recv-Q     Send-Q          Local Address:Port          Peer Address:Port     Process
udp      UNCONN    0          0                  127.0.0.54:53                 0.0.0.0:*          users:(("systemd-resolve",pid=128737,fd=16))
udp      UNCONN    0          0               127.0.0.53%lo:53                 0.0.0.0:*          users:(("systemd-resolve",pid=128737,fd=14))
udp      UNCONN    0          0             172.16.0.4%eth0:68                 0.0.0.0:*          users:(("systemd-network",pid=128580,fd=22))
udp      UNCONN    0          0                   127.0.0.1:323                0.0.0.0:*          users:(("chronyd",pid=1086,fd=6))
udp      UNCONN    0          0                       [::1]:323                   [::]:*          users:(("chronyd",pid=1086,fd=7))
tcp      LISTEN    0          4096               127.0.0.54:53                 0.0.0.0:*          users:(("systemd-resolve",pid=128737,fd=17))
tcp      LISTEN    0          100                   0.0.0.0:25                 0.0.0.0:*          users:(("master",pid=132927,fd=13))
tcp      LISTEN    0          10                    0.0.0.0:23                 0.0.0.0:*          users:(("inetutils-inetd",pid=147481,fd=4))
tcp      LISTEN    0          4096                  0.0.0.0:22                 0.0.0.0:*          users:(("sshd",pid=112304,fd=3),("systemd",pid=1,fd=116))
tcp      LISTEN    0          4096            127.0.0.53%lo:53                 0.0.0.0:*          users:(("systemd-resolve",pid=128737,fd=15))
tcp      LISTEN    0          100                      [::]:25                    [::]:*          users:(("master",pid=132927,fd=14))
tcp      LISTEN    0          4096                     [::]:22                    [::]:*          users:(("sshd",pid=112304,fd=4),("systemd",pid=1,fd=117))
```

```
None
sudo ss -tulpn
```

Service/Port	Keep or Disable	Reasoning
systemd-resolved — DNS, port 53 on 127.0.0.53/54	Keep	Provides local DNS resolution. The addresses are loopback-only (`127.0.0.x`), so this is not exposed externally.
systemd-networkd — DHCP client, UDP 68	Keep	Handles network configuration/DHCP for the VM. The server needs this for network connectivity.
chronyd — NTP, UDP 323 on 127.0.0.1	Keep	Maintains accurate system time, which is important for logs, auditing, authentication, and certificates.
Postfix — SMTP, TCP 25	Disable	Provides SMTP mail service. A banking web/file server does not normally need to act as a mail server, so this is unnecessary exposure.
inetutils-inetd / Telnet — TCP 23	Keep	
sshd — SSH, TCP 22	Keep	Provides secure remote administration. This is needed to manage the server remotely.
Systemd — TCP 22	Keep	This appears alongside SSH because systemd is involved in managing/socket activation for the SSH service. It is not a separate network service you would disable here.
ModemManager.service — No Port Listed	Disable	ModemManager was found running in the systemctl service list, but it does not appear in ss -tulpn because it is not currently listening on a TCP or UDP port. An Azure banking web/file server does not need cellular modem management, so the service is unnecessary and can be disabled to reduce the system's attack surface.

Disable:

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo systemctl disable --now ModemManager.service
[sudo] password for student:
Removed "/etc/systemd/system/multi-user.target.wants/ModemManager.service".
Removed "/etc/systemd/system/dbus-org.freedesktop.ModemManager1.service".
student@LinuxTargetVM3:~$ sudo systemctl disable --now postfix.service
Synchronizing state of postfix.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install disable postfix
Removed "/etc/systemd/system/multi-user.target.wants/postfix.service".
```

None

```
sudo systemctl disable --now <service-name>
```

Confirmation:

None

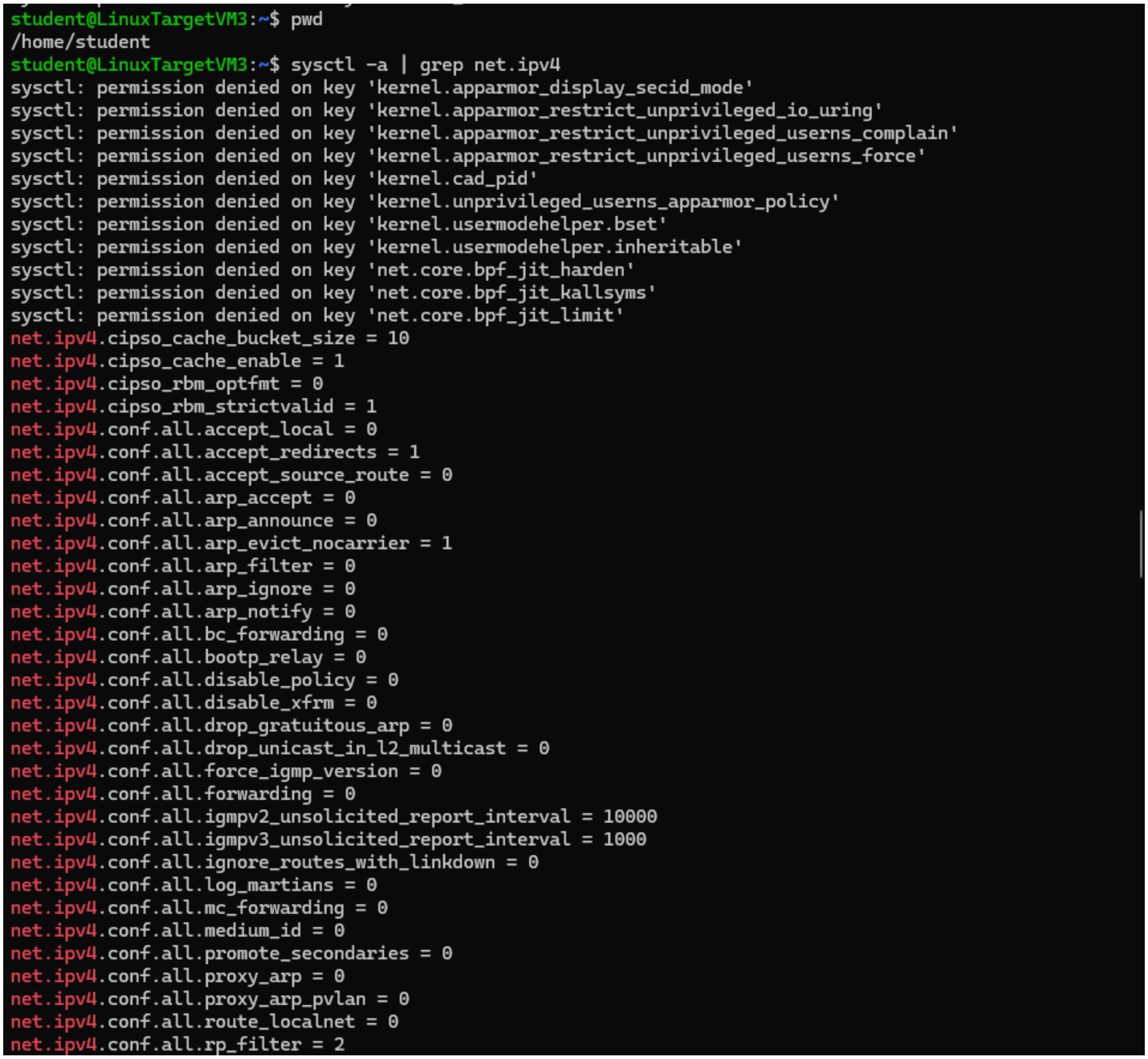
```
sudo ss -tulpn
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo ss -tulpn
Netid    State     Recv-Q    Send-Q         Local Address:Port       Peer Address:Port    Process
udp      UNCONN    0         0                 127.0.0.54:53              0.0.0.0:*        users:(("systemd-resolve",pid=128737,fd=16))
udp      UNCONN    0         0              127.0.0.53%lo:53              0.0.0.0:*        users:(("systemd-resolve",pid=128737,fd=14))
udp      UNCONN    0         0            172.16.0.4%eth0:68              0.0.0.0:*        users:(("systemd-network",pid=128580,fd=22))
udp      UNCONN    0         0                  127.0.0.1:323             0.0.0.0:*        users:(("chronyd",pid=1086,fd=6))
udp      UNCONN    0         0                      [::1]:323                [::]:*        users:(("chronyd",pid=1086,fd=7))
tcp      LISTEN    0         4096              127.0.0.54:53              0.0.0.0:*        users:(("systemd-resolve",pid=128737,fd=17))
tcp      LISTEN    0         10                   0.0.0.0:23              0.0.0.0:*        users:(("inetutils-inetd",pid=147481,fd=4))
tcp      LISTEN    0         4096                 0.0.0.0:22              0.0.0.0:*        users:(("sshd",pid=112304,fd=3),("systemd",pid=1,fd=113))
tcp      LISTEN    0         4096           127.0.0.53%lo:53              0.0.0.0:*        users:(("systemd-resolve",pid=128737,fd=15))
tcp      LISTEN    0         4096                    [::]:22                [::]:*        users:(("sshd",pid=112304,fd=4),("systemd",pid=1,fd=115))
```

```
student@LinuxTargetVM3:~$ systemctl status ModemManager.service postfix.service
○ ModemManager.service - Modem Manager
     Loaded: loaded (/usr/lib/systemd/system/ModemManager.service; disabled; preset: enabled)
     Active: inactive (dead)

Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: ModemManager.service: Deactivated successfully.
Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: Stopped ModemManager.service - Modem Manager.
Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: Starting ModemManager.service - Modem Manager...
Aug 11 06:04:33 LinuxTargetVM3 ModemManager[132545]: <msg> ModemManager (version 1.23.4) starting in system bus...
Aug 11 06:04:33 LinuxTargetVM3 systemd[1]: Started ModemManager.service - Modem Manager.
Aug 11 18:52:02 LinuxTargetVM3 ModemManager[132545]: <msg> caught signal, shutting down...
Aug 11 18:52:02 LinuxTargetVM3 systemd[1]: Stopping ModemManager.service - Modem Manager...
Aug 11 18:52:02 LinuxTargetVM3 ModemManager[132545]: <msg> ModemManager is shut down
Aug 11 18:52:02 LinuxTargetVM3 systemd[1]: ModemManager.service: Deactivated successfully.
Aug 11 18:52:02 LinuxTargetVM3 systemd[1]: Stopped ModemManager.service - Modem Manager.

○ postfix.service - Postfix Mail Transport Agent
     Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; preset: enabled)
     Active: inactive (dead)
       Docs: man:postfix(1)

Aug 10 18:21:31 LinuxTargetVM3 systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Aug 10 18:21:31 LinuxTargetVM3 systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
Aug 11 18:52:12 LinuxTargetVM3 systemd[1]: postfix.service: Deactivated successfully.
Aug 11 18:52:12 LinuxTargetVM3 systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
```

Part 3: Kernel Hardening with sysctl

1. Review current network-related kernel settings:



```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sysctl -a | grep net.ipv4
sysctl: permission denied on key 'kernel.apparmor_display_secid_mode'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_io_uring'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_userns_complain'
sysctl: permission denied on key 'kernel.apparmor_restrict_unprivileged_userns_force'
sysctl: permission denied on key 'kernel.cad_pid'
sysctl: permission denied on key 'kernel.unprivileged_userns_apparmor_policy'
sysctl: permission denied on key 'kernel.usermodehelper.bset'
sysctl: permission denied on key 'kernel.usermodehelper.inheritable'
sysctl: permission denied on key 'net.core.bpf_jit_harden'
sysctl: permission denied on key 'net.core.bpf_jit_kallsyms'
sysctl: permission denied on key 'net.core.bpf_jit_limit'
net.ipv4.cipso_cache_bucket_size = 10
net.ipv4.cipso_cache_enable = 1
net.ipv4.cipso_rbm_optfmt = 0
net.ipv4.cipso_rbm_strictvalid = 1
net.ipv4.conf.all.accept_local = 0
net.ipv4.conf.all.accept_redirects = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.all.arp_accept = 0
net.ipv4.conf.all.arp_announce = 0
net.ipv4.conf.all.arp_evict_nocarrier = 1
net.ipv4.conf.all.arp_filter = 0
net.ipv4.conf.all.arp_ignore = 0
net.ipv4.conf.all.arp_notify = 0
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.bootp_relay = 0
net.ipv4.conf.all.disable_policy = 0
net.ipv4.conf.all.disable_xfrm = 0
net.ipv4.conf.all.drop_gratuitous_arp = 0
net.ipv4.conf.all.drop_unicast_in_l2_multicast = 0
net.ipv4.conf.all.force_igmp_version = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.igmpv2_unsolicited_report_interval = 10000
net.ipv4.conf.all.igmpv3_unsolicited_report_interval = 1000
net.ipv4.conf.all.ignore_routes_with_linkdown = 0
net.ipv4.conf.all.log_martians = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.all.medium_id = 0
net.ipv4.conf.all.promote_secondaries = 0
net.ipv4.conf.all.proxy_arp = 0
net.ipv4.conf.all.proxy_arp_pvlan = 0
net.ipv4.conf.all.route_localnet = 0
net.ipv4.conf.all.rp_filter = 2
```

None

```
sysctl -a | grep net.ipv4
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ sudo nano /etc/sysctl.d/99-hardening.conf
student@LinuxTargetVM3:~$
```

```None
sudo nano /etc/sysctl.d/99-hardening.conf
```

Lines before:

```
net.ipv4.tcp_syncookies = 1
```

```
net.ipv4.ip_forward = 0
```

```
net.ipv4.icmp_echo_ignore_broadcasts = 1
```

```
net.ipv4.conf.all.accept_source_route = 0
```

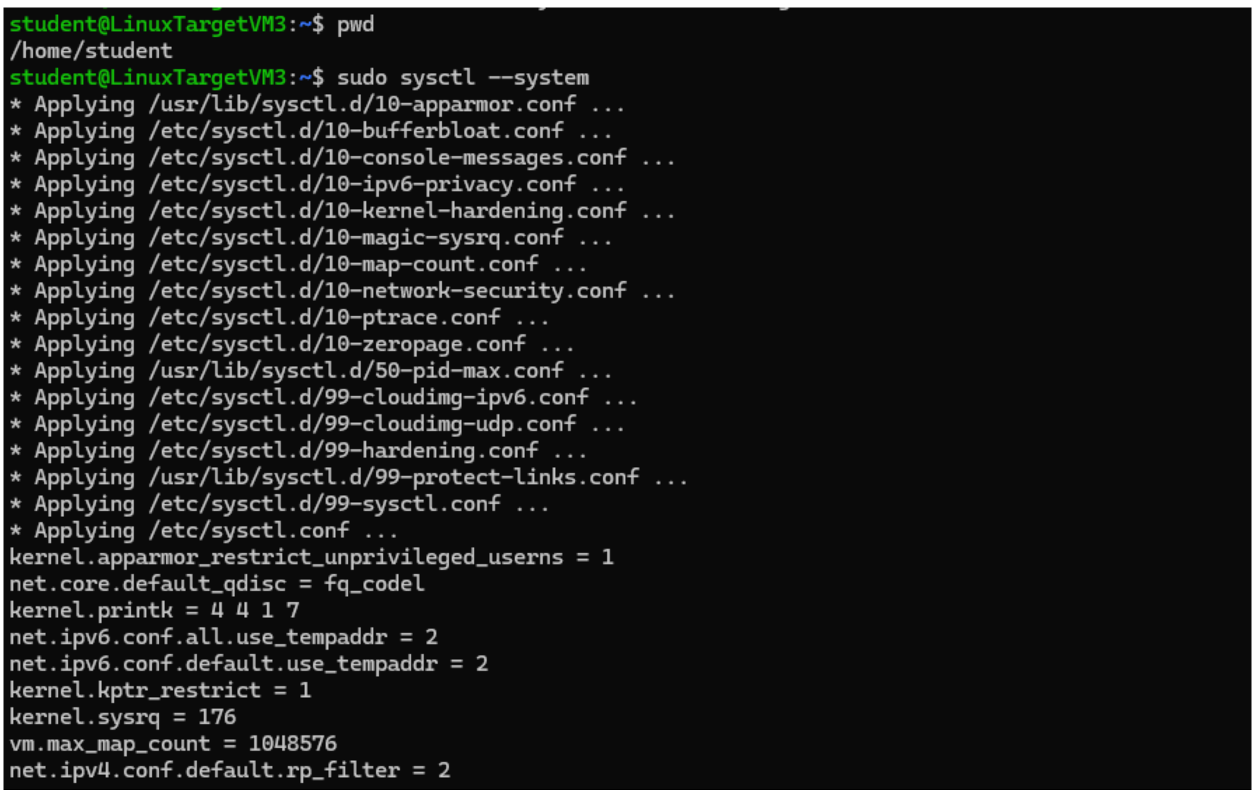
```
net.ipv4.conf.default.accept_source_route = 1
```

Add these lines:

```
  GNU nano 7.2                          /etc/sysctl.d/99-hardening.conf
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0

                                   [ Wrote 5 lines ]
^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

```None
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
```

Apply the changes:




```None
sudo sysctl --system
```

```
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
fs.protected_fifos = 1
fs.protected_hardlinks = 1
```

None
```
sysctl net.ipv4.ip_forward
sysctl net.ipv4.tcp_syncookies
```

```
student@LinuxTargetVM3:~$ pwd
/home/student
student@LinuxTargetVM3:~$ cat /etc/sysctl.d/99-hardening.conf
net.ipv4.ip_forward = 0
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
```

This is what the /etc/sysctl.d/99-hardening.conf file includes.

5. Hardening Summary

The Linux host was hardened by applying all available system and security updates to reduce exposure to any known vulnerabilities. Unattended upgrades were also configured so future security patches could be applied automatically.

We reviewed active services and listening ports and determined that Postfix and ModemManager were deemed unnecessary for the banking web/file server role, so both of those services were disabled to reduce the attack surface of the system. Telnet was identified as insecure but was intentionally left enabled since it was required for a later portion of the lab.

6. Reflection Questions

1. Which change today felt riskiest to make on a production-bound host, and how did you validate it didn't break anything?

The riskiest change made today was most likely the one involving telnet. As a weak application is can cause issues just by enabling it. Even though it was not totally enforced and verified it still caused issues that would need to be validated that it did not break the machine.

2. Why might disabling a service be a better fix than just blocking its port with a firewall rule? Are there cases where you'd want both?

Disabling a service reduces the attack surface as well as exposure. A firewall can fail or get bypassed so it adds another layer of security also known as defense in depth. In most real world settings both are the common setup for services you would need because of the added layers of defense. An example would be if a public web server has to be reachable due to legacy software. The firewall can be configured to allow only http and https and deny everything else.

3. How does mandatory access control (AppArmor) protect a host differently than the least-privilege user/file permissions work from Week 1

Least-privilege access permissions are used to control what users and groups are allowed to access on the system. AppArmor is able to go a step further by restricting what specific applications are allowed to do. Even if a user has permission to access something, AppArmor would still block them from accessing it. All this does is add another layer of security that helps limit any damage if a specific application is compromised. This goes into accordance of defense in depth by limiting both user privileges and application behavior.